

4.4 Addressing the Research Questions

This study aimed to develop and evaluate a system that predicts employment opportunities using Random Forest algorithm with a web application. The findings address each of the research questions posed at the beginning of the study:

4.4.1 How can Random Forest machine learning techniques effectively predict employment opportunities?

The results demonstrate that Random Forest techniques can effectively predict employment opportunities through a multi-staged approach. The model achieved an overall accuracy of 86.3% on the independent test dataset with balanced performance across precision (84.9%) and recall (83.5%) metrics, successfully classifying across 639 distinct job categories. The effectiveness of the Random Forest approach is attributable to several factors:

First, the comprehensive preprocessing and feature engineering transformed unstructured text data into meaningful numerical representations. The ordinal encoding of education levels preserved hierarchical relationships, while the binary representation of skills and numerical extraction from experience text created structured features that enabled the algorithm to identify patterns.

Second, the ensemble nature of Random Forest mitigated overfitting, as evidenced by the remarkably consistent cross-validation results (standard deviation $\leq 0.55\%$ across metrics). This stability indicates that the model generalizes well across different data subsets rather than memorizing training examples.

Third, the stratified sampling approach ensured balanced representation of job categories during training and evaluation, preventing bias toward common roles. This was crucial given the relatively balanced distribution of job roles in the dataset, where even the most common roles represented less than 0.2% of the data.

Finally, hyperparameter optimization identified settings that balanced model complexity with generalization capability. The selected configuration with 200 estimators provided sufficient ensemble diversity to capture complex patterns, while the controlled depth of 20 prevented overfitting to training data noise.

4.4.2 What features are most influential in determining employment potential?

The feature importance analysis revealed clear patterns regarding the most influential factors in determining employment potential:

Among technical skills, programming languages and frameworks consistently emerged as the most influential predictors, with Python ranking as the most important feature overall, followed by Java and SQL. This finding aligns with the increasing demand for programming skills across diverse professional domains beyond traditional software development.

Experience-related features demonstrated substantial predictive power, with total years of experience ranking as the fourth most important feature. This confirms the significant role of professional tenure in determining appropriate job roles, reflecting the career progression patterns observed in the exploratory data analysis.

Education level was important but ranked lower than many technical skills in the overall importance hierarchy. This suggests that in numerous fields, specialized skills may outweigh formal education credentials in determining suitable employment opportunities—though this pattern varied across domains. In healthcare and research roles, education level retained high importance, reflecting the formal qualification requirements in these fields.

The SHAP analysis revealed complex interactions between features, with certain skill combinations demonstrating synergistic effects. For instance, the combination of Python and machine learning skills had greater predictive impact than the sum of their individual contributions when classifying data scientist roles, highlighting the importance of complementary skill sets.

4.4.3 How can a web application enhance the accessibility and usability of employment predictions?

The web application developed in this study enhanced the accessibility and usability of employment predictions in several key ways:

First, the application provided multiple input options, allowing users to upload resumes in various formats (PDF, DOCX, image) or manually enter their information through structured forms. This flexibility accommodated different user preferences and improved accessibility, achieving a 94.2% accuracy rate for information extraction across formats.

Second, the intuitive user interface design (rated 4.6/5 by test users) simplified the interaction process, making sophisticated machine learning predictions accessible to users regardless of technical background. The modular component structure with 84 distinct UI components organized into 8 main page components provided a cohesive user experience from profile submission to results presentation.

Third, the client-server architecture made computationally intensive processes transparent to users. The serverless function deployment via Vercel's edge network ensured low-latency access globally, with prediction response times under 500ms for typical user profiles, making the system responsive and user-friendly.

Fourth, the integration of real-time job listings related to predicted roles transformed abstract predictions into actionable opportunities. This feature achieved an 87.4% relevance rate for top search results, providing immediate practical value to users without requiring them to conduct separate job searches.

Finally, the Progressive Web Application design ensured accessibility across devices and network conditions, making career guidance available to users with limited connectivity or computing resources, an important consideration in the Philippine context.

4.4.4 How can the developed web application provide personalized career recommendations?

The web application provides personalized career recommendations through a sophisticated hybrid approach that combines traditional machine learning with contextual AI understanding:

The system delivers personalized recommendations by first extracting individual profile details through advanced resume parsing (achieving 94.2% overall accuracy) or structured manual input forms. This creates a detailed understanding of each user's unique combination of education, skills, and experience.

The Random Forest model analyzes these features to generate the top 4-7 most relevant job roles based on patterns identified from 76,294 professional profiles. The number of recommendations is dynamically adjusted based on confidence scores, providing more options when multiple roles have similar probability scores.

The integration of Large Language Models through the Vercel AI SDK generates tailored insights that go beyond simple role matching. This includes strengths assessment, personalized skill development recommendations, and career advice contextualized to both the user's profile and the predicted roles (achieving a human-rated relevance score of 4.6/5).

The system also implements a bias detection and correction mechanism that analyzes prediction patterns across different demographic groups to ensure fairness in the recommendations.

User testing revealed high satisfaction with the personalization aspects of the system, particularly the skill development recommendations (rated 4.7/5) and the contextualized explanations connecting user skills to job requirements. As one user commented: "The system provided job recommendations I hadn't considered but that aligned well with my skillset."

4.5 Challenges Encountered and Lessons Learned

The development of the employment opportunity prediction system presented several significant challenges that required innovative solutions. Reflecting on these challenges provides valuable insights for future research and development in this domain.

4.5.1 Complexity and Variability of Resume Parsing

Challenge: Extracting accurate and consistent structured data (education, skills, experience) from diverse resume formats proved challenging. Resumes vary widely in structure, formatting, terminology, and level of detail, making automated information extraction particularly difficult.

Solution Approach: A hybrid approach combining rule-based extraction with machine learning classification was implemented. Named entity recognition (NER) using spaCy identified key entities such as organization names, job titles, and dates, supplemented with custom pattern matching rules for education credentials and domain-specific terminology. Additionally, the system implemented an interactive validation step where users could review and modify automatically extracted information, ensuring accuracy while reducing the need for perfect automatic extraction.

Lesson Learned: While sophisticated NLP techniques can achieve impressive results (94.2% overall accuracy in our implementation), perfect automatic extraction remains elusive due to the inherent variability of human-created documents. Designing systems that combine automated processing with human validation creates a more robust solution than attempting to solve the problem through automation alone.

4.5.2 Integration of Python-based Machine Learning with Next.js

Challenge: Integrating a Python-trained Random Forest model into a predominantly TypeScript/Next.js environment presented significant technical hurdles. The different ecosystems have distinct runtime environments, data structures, and serialization formats.

Solution Approach: The solution involved careful handling of model serialization and deserialization using joblib to create a compact binary representation of the trained model. A serverless function was implemented to load and utilize the Python model within Next.js API routes. This required custom middleware that bridged the TypeScript request handling with Python model inference. Additionally, a preprocessing pipeline that exactly matched the training preprocessing steps was reimplemented in the production environment.

Lesson Learned: Cross-language integration in machine learning applications requires meticulous attention to preprocessing consistency and data transformation. Establishing clear interfaces between components developed in different languages, with comprehensive testing to ensure equivalence, is essential for maintaining prediction accuracy in production environments.

4.5.3 Balancing Model Complexity and Performance Requirements

Challenge: Deploying a Random Forest model with 200 estimators and a large feature space in a serverless environment introduced potential latency concerns, especially when combined with LLM interaction for insight generation.

Solution Approach: The system architecture was optimized through several techniques: model pruning to reduce memory footprint while maintaining accuracy, intelligent caching of model loading to avoid repeated initialization costs, asynchronous processing where appropriate, and selective feature computation that only calculated necessary features based on the user's input. Additionally, the deployment architecture utilized Vercel's edge network to minimize network latency.

Lesson Learned: Production machine learning systems require careful consideration of the trade-offs between model complexity, accuracy, and performance constraints. Incremental optimization through profiling and targeted improvements can significantly enhance system responsiveness without sacrificing prediction quality.

4.5.4 Effective Integration of LLM for Contextual Insights

Challenge: While the Random Forest model provided accurate job role predictions, generating meaningful, contextual explanations and career advice required integration with Large Language Models. This presented challenges in prompt engineering, response quality control, and ensuring consistency with the statistical predictions.

Solution Approach: A structured prompt template system was developed that incorporated both the Random Forest predictions and user profile data. The prompts were carefully engineered through iterative testing to guide the LLM toward producing insights relevant to the specific user and predicted roles. Additionally, post-processing of LLM outputs ensured consistent formatting and filtering of potentially problematic content before presentation to users.

Lesson Learned: Effective LLM integration requires treating prompt engineering as a formal design process with systematic testing and refinement. The quality of AI-generated insights depends heavily on how information is structured and presented to the model. Building robust interfaces between statistical models and generative AI components enables leveraging the strengths of both approaches.

4.6 Comparison with Related Literature

This section compares the findings of the current study with previous research on employment prediction, particularly in the Philippine context, highlighting areas of agreement, extension, and divergence.

4.6.1 Random Forest Performance for Employment Prediction

The current study achieved an overall accuracy of 86.3% with the Random Forest model for job role prediction across 639 categories. This performance exceeds the 82% accuracy reported by Mercado and Lee (2023) in their Random Forest model specifically for Philippine job market analysis. The improved performance may be attributed to several factors: our more extensive feature engineering process, particularly the binary encoding of 34 distinct skills; the optimization of hyperparameters through grid search; and the larger training dataset (76,294 profiles compared to their reported 25,000 samples).

Similar to the findings of Delos Santos and Kumar (2023), who applied Random Forest techniques to analyze employment opportunities across different Philippine regions, our research confirmed the algorithm's effectiveness in capturing complex relationships between candidate attributes and suitable roles. However, while their study focused primarily on geographic variations in employment patterns, our research expanded the scope to include detailed analysis of the interplay between education, skills, and experience across 639 job categories.

4.6.2 Feature Importance and Skill Demands

Our feature importance analysis revealed that programming languages (Python, Java, SQL) and technical skills ranked as the most influential predictors, followed by years of experience, with formal education ranking lower for many fields. This partially aligns with De Guzman and Tan's (2023) analysis of job postings from major Philippine job boards, which similarly identified technical skills as critical, though they placed greater emphasis on communication skills than our model did.

Ramirez et al. (2022), who tracked changing skill requirements across various sectors in the Philippines from 2019-2023, identified an increasing emphasis on digital competencies across industries—a trend confirmed by our finding that programming skills were highly predictive even for roles outside traditional IT. However, while their qualitative analysis suggested communication skills were becoming more important over time, our quantitative model ranked these lower in predictive importance, though still present in the top 10 features.

The synergistic effect between complementary skills (such as Python and machine learning) identified through our SHAP analysis represents a novel finding not explicitly addressed in previous Philippine employment studies, suggesting that skill combinations may be more important than individual skills in determining employment opportunities—an insight that extends beyond the individual skill analysis common in earlier research.

4.6.3 Regional and Sectoral Variations

Yu and Trinidad (2024) developed a regional employment opportunity index using Random Forest algorithms and found significant variations in employment patterns across Philippine regions. While our study did not explicitly model geographic variations, the variance in model performance across job domains (92.3% precision for IT & Software Development versus 78.3% for Administrative & Support) suggests that prediction effectiveness may indeed vary by sector in ways that could correlate with regional concentrations of industries.

Ignacio et al. (2023) developed industry-specific predictive models for Philippine employment and reported improved performance with sector-specific models compared to general predictions. Our finding that technical fields demonstrated superior precision in predictions (IT: 92.3%, Data Science: 89.7%) compared to other domains partially validates their sector-specific approach, suggesting that some industries may benefit from specialized models.

4.6.4 Technology Integration and User Experience

Angeles and Martinez (2024) studied how Philippine companies are utilizing AI-driven recruitment tools and reported a 30% improvement in matching efficiency with data-driven approaches. Our user testing results extend these findings from the employer perspective to the job seeker side, demonstrating that AI-assisted career guidance can achieve high user satisfaction (4.4/5 overall) and provide relevant recommendations that users might not have otherwise considered.

The hybrid approach combining statistical machine learning with LLM-generated insights in our system represents an advancement beyond the purely statistical methods described in most Philippine employment prediction literature. Villanueva et al. (2023), while examining digital skills matching platforms, primarily focused on algorithmic matching without the contextual explanation capabilities our system provides through LLM integration—a feature that received particularly high user ratings (4.6/5) in our testing.

4.7 Limitations Revisited

The development and evaluation of the employment opportunity prediction system revealed several limitations that warrant acknowledgment and exploration of their potential impact on the findings and system performance.

4.7.1 Dataset Representation and Coverage

As identified in Chapter 1, the system relies exclusively on the Professional Profiles dataset from Hugging Face, containing 76,294 profiles. While this dataset provided a diverse collection of professional backgrounds, it may not fully represent the unique characteristics of the Philippine labor market, particularly for region-specific employment patterns or industry-specific practices. The dataset's global nature means that certain job categories, skills, or education patterns more common in the Philippines might be underrepresented.

This limitation potentially impacts the accuracy of predictions for distinctly Philippine job roles or emerging positions in the local market. Future iterations could benefit from supplementing the training data with Philippines-specific professional profiles to better capture local employment nuances.

4.7.2 Handling of Unconventional or Incomplete Inputs

The system demonstrated challenges in processing highly unusual or incomplete user inputs, as anticipated in the initial limitations. During user testing, profiles with unconventional career paths or non-standard skill descriptions received the lowest satisfaction rating (3.8/5). While the resume parsing component achieved a 94.2% overall accuracy, performance declined for resumes with highly design-oriented layouts or unconventional structures.

This limitation affects the system's accessibility for users with atypical career trajectories or those unable to provide comprehensive information. The manual input option partially mitigates this issue by allowing direct entry of information, but the underlying prediction model may still struggle with unusual combinations of education, skills, and experience that weren't well-represented in the training data.

4.7.3 Historical Data and Market Dynamics

The prediction model's reliance on historical data limits its ability to anticipate rapid or sudden changes in the job market—a limitation that became particularly evident for emerging roles in technology. While the integration of LLMs helped mitigate this issue by incorporating more recent contextual understanding of job markets, the fundamental patterns learned by the Random Forest model remain anchored to the time period represented in the training data.

This temporal limitation may reduce the system's effectiveness during periods of significant labor market disruption or for emerging job categories. The web application's capacity to retrieve current job postings provides a partial remedy by connecting predictions to real-time opportunities, but the underlying model recommendations may lag behind rapid market evolutions.

4.7.4 Scope of System Functionality

As specified in the initial limitations, the system's functionality focuses on providing predictions and insights rather than facilitating direct job applications or employer connections. User feedback indicated interest in additional features beyond the current scope, such as direct application capabilities (mentioned by 47% of test users) and employer matching (requested by 35%).

While the current implementation successfully meets its core objectives of providing accurate predictions and personalized insights, this limitation affects the system's utility as a comprehensive employment solution. Users must still navigate to external platforms to act on the system's recommendations, creating potential friction in the job search process.

4.7.5 Multilingual and Cultural Adaptation

An additional limitation that emerged during implementation was the system's limited support for non-English resumes and profiles. The resume parsing component and Random Forest model were primarily optimized for English-language inputs, potentially affecting accessibility for users more comfortable providing information in Filipino or other Philippine languages.

This language limitation may impact the system's inclusivity and reach within the diverse Philippine linguistic landscape, particularly in regions where English proficiency varies. Future development should consider multilingual support to better serve the national context.